

YUCHEN WANG

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SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, HTML/CSS, Golang

Cloud & Ops: AWS (Lambda, EC2, S3, DynamoDB, RDS, CloudWatch, CloudFormation), AWS SageMaker, Bedrock, GCP ONNX/ONNX Runtime, TorchScript, TorchServe, FastAPI, GCP, Docker, Kubernetes, GitHub Actions, REST APIs, CI/CD

ML & AI: PyTorch, TensorFlow, Keras, Scikit-Learn, LangChain, LangGraph, LangSmith, Hugging Face, Gymnasium, Ollama, Stable-Baselines3, FAISS, Pinecone, Chroma, NLTK

Certification & Methods: AWS Cloud Practitioner (CLF-C02), Object-Oriented Design, SDLC (Agile/Scrum/V-Model)

EXPERIENCE

Software Development Engineer Intern, Amazon Web Services – Seattle, Washington May 2025 – Aug 2025

- Delivered two production-ready automations with internal AWS tooling deployed in multiple deployment stages; collaborated cross-team (engineering, security, docs) for multi-region adoption of automation in 40+ AWS regions
- Automated Canary Vulnerability Scanner by integrating AWS Inspector, CloudWatch, CloudFormation to auto-detect high severity package CVEs and page on-call engineers; drove high-severity tickets on package to 0 in the next release
- Created an one-command package deployment automation that generated versioned docs, opened code revisions, and filed tickets to owning teams with agentic setup process using Amazon Q; saving 300+ operation hours per year

Business & Software Intern, Desai Business Accelerator – Ann Arbor, Michigan May 2024 – Aug 2024

- Designed prototype of an AI translation software using React.js, Express.js, and TensorFlow to facilitate rapid language processing and deployable cloud services, tested by 100 users to for UX and scalability
- Built and deployed WhatsApp Business integration using Node.js, Express.js and WhatsApp API, enabling seamless client communication, ensuring reliability with Jenkins for CI/CD and successfully facilitating 30+ deals
- Implemented frontend and backend of game catalog for startup's public launch using Angular and Express.js, connected to MySQL and Amazon S3 bucket, holding 30+ games in 45+ subcategories on release

Computer Science Instructional Aide, University of Michigan – Ann Arbor, Michigan Aug 2024 – Dec 2024

- Collaborated with University Professors weekly on lecture material on core mathematical proofs and logic in a course with 600+ students, contributing to a 13% improvement in student engagement measured by surveys
- Led weekly discussion sessions for 49 students, delivering interactive lessons on key course concepts while maintaining direct communication between administration and students, holding a 95% satisfaction rate
- Authored 12 weekly homework and check-in questions directly with University of Michigan professors tailored to address key learning objectives, leading to a 25% increase in student comprehension and resource engagement

PROJECTS

ML Paper Companion github.com/wangychn/ai-paper-chatbot-fullstack

- Designed and implemented full-stack ml paper research assistant via LangGraph RAG pipeline backend and React + Tailwind frontend, integrated LangSmith for tracing and observability, and deployed backend as LangGraph server
- Built Python backend with LangGraph and FastAPI to support tool-augmented LLM workflows, including custom arXiv search tools, embeddings ingestion, and retrieval-grounded generation with streaming SSE responses
- Implemented responsive chat interface frontend in React with SSE hooks for real-time model outputs and structured paper results; implemented modular components (MessageList, PaperCard, MessageInput) for clean UX

Cart-Pole DQN Experiment github.com/wangychn/CartPole-DQN

- Developed a PyTorch Deep Q-Network (2×64 MLP) to solve the classic Cart-Pole control task, mapping the 4-D state to Q-values for two actions and training with Adam to minimize mean-squared Bellman error
- Achieved “solved” performance; built a reproducible pipeline with configurable scripts, live training/test curves, ϵ -decay plots, and periodic evaluation rollouts and exported Cart-Pole policy to ONNX for web-based demo
- Implemented stabilization and exploration strategies—target network synchronization, experience replay (70k capacity, batch 128), ϵ -greedy schedule (ϵ 1.0→0.01, 0.997 decay/episode), and exponential learning-rate decay

EDUCATION

University of Michigan - Ann Arbor – B.S.

Computer Science

Relevant Coursework: Machine Learning (EECS 445), Data Structures & Algorithms (EECS 281), Linear Algebra (MATH 214)

2027

GPA: 3.7/4.0